

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Title - Predictive Maintenance of Industrial Machinery (PS39)

Domain of Project – Manufacturing / Industrial Engineering

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Mohammad Basheer Shabbir Shareef		Basheershareef2005@gmail.com	7981841214

Problem Statement -

Problem Statement : Predictive Maintenance of Industrial Machinery

The Challenge Industrial fleets face catastrophic failures due to wear and thermal stress, but maintenance teams often struggle to anticipate these events before they occur. Information about machine health is buried within extreme volumes of sensor telemetry data. Without **proactive, real-time intelligence**, many organizations rely on reactive maintenance, leading to costly downtime and inefficient operations.

The Objective Build an **intelligent web application** using machine learning that delivers **real-time, proactive industrial asset monitoring**. The solution should:

- **Predict Failure Types** – Anticipate specific anomalies including Tool Wear, Heat Dissipation, Power Failure, and Overstrain.
- **Live Sensor Input** – Accept and process real-time telemetry such as Rotational Speed, Torque, Air Temperature, and Process Temperature.
- **Diagnostic Engine** – Combine an IBM Watsonx AutoAI Machine Learning model (Snap Random Forest) with rule-based mechanical overrides.
- **Mitigation Protocols** – Provide actionable, step-by-step mitigation strategies based on the specific anomaly detected.
- **Visualization Dashboard** – Display real-time telemetry, model confidence levels, and safety warnings in an intuitive Streamlit interface.

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used IBM watsonx.ai (AutoAI) + Streamlit + Python

Proposed Solution -

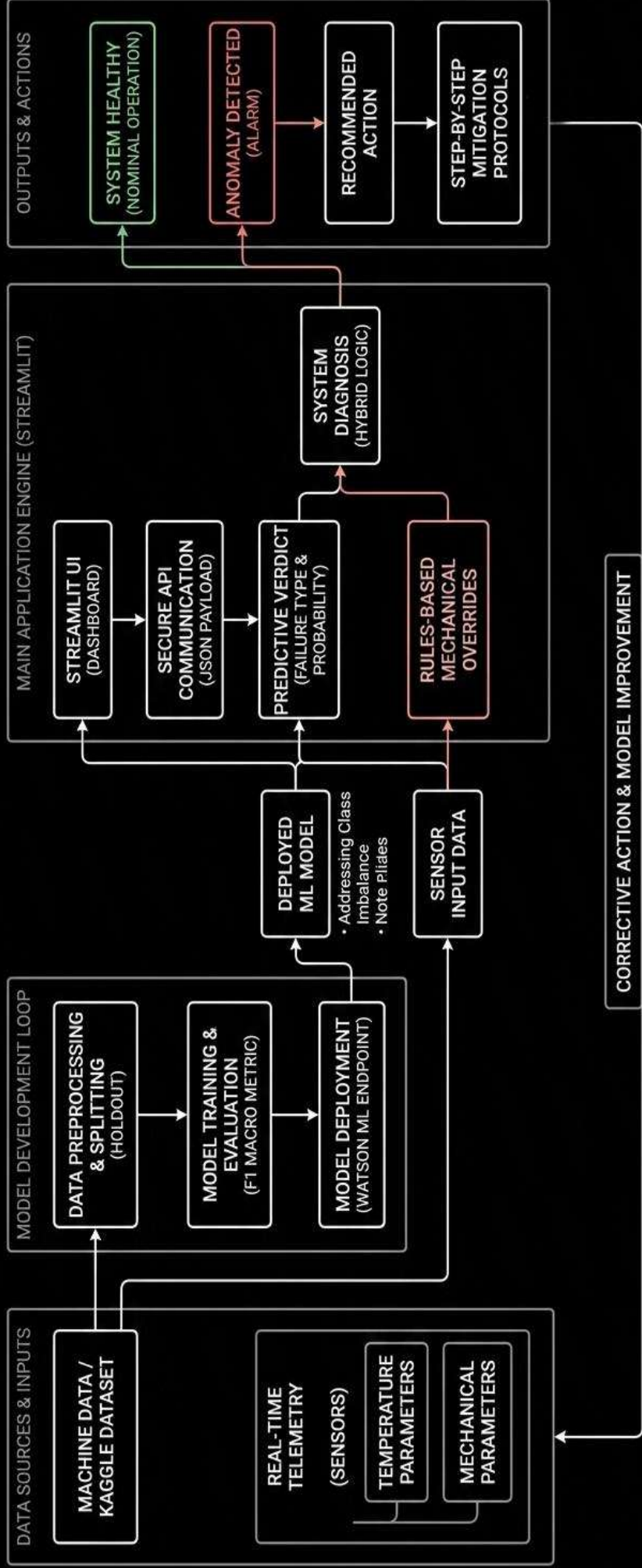
- **Machine Learning (AI) Insights:**
- **Data Analysis:** Analyzes real-time sensor data—like temperature, rotational speed, torque, and tool wear—collected from your machines.
- **Failure Classification:** Uses a trained machine learning model conceptually optimized for tasks like F1 Macro-averaging (to handle rare failure cases effectively). This model conceptualizes predictions on *when* a failure might occur and *what type* of failure it is likely to be (e.g., Tool Wear, Heat Dissipation, Power Failure, or Overstrain).
- **Secure API:** Accesses these predictions through a deployed endpoint, possibly conceptualizing standard cloud or edge services.
- **Rule-Based Safety Logic:**
- **Override Protocols:** Operates a concurrent rules engine containing hardcoded mechanical safety thresholds (e.g., maximum permissible temperature, extreme speed-to-torque ratios).
- **Immediate Action:** If any of these critical limits are conceptually breached, the rules-based system can immediately trigger alarms or shutdown protocols, prioritizing safety even if the ML model is still processing or has lower confidence.

Streamlit component Used -

- 1) **Streamlit Input Panel** - The interactive sidebar where operators manually enter or adjust real-time machine telemetry, such as rotational speed, torque, and temperatures.
- 2) **IBM watsonx AutoAI Engine** - The cloud backend that receives the sensor data and processes it using optimized Machine Learning pipelines to generate failure predictions.
- 3) **Snap Random Forest Classifier** - The high-performance machine learning algorithm selected by AutoAI. Fast and efficient for real-time multi-class failure classification.
- 4) **Streamlit Diagnostic Output** - The main visual dashboard that displays the AI-generated results, including failure probability charts, risk levels, and step-by-step mitigation protocols.
- 5) **Rule-Based Safety Engine** - Hardcoded mechanical logic within the app that overrides the AI to trigger immediate warnings if critical physical thresholds (like extreme heat) are breached.
- 6) **Streamlit Secrets Vault** - The secure configuration system (`secrets.toml`) that safely stores and injects your IBM Cloud API key to connect the public frontend to the private backend.

Blueprint

WORKFLOW / ARCHITECTURE - PREDICTIVE MAINTENANCE SYSTEM



Architecture blueprint

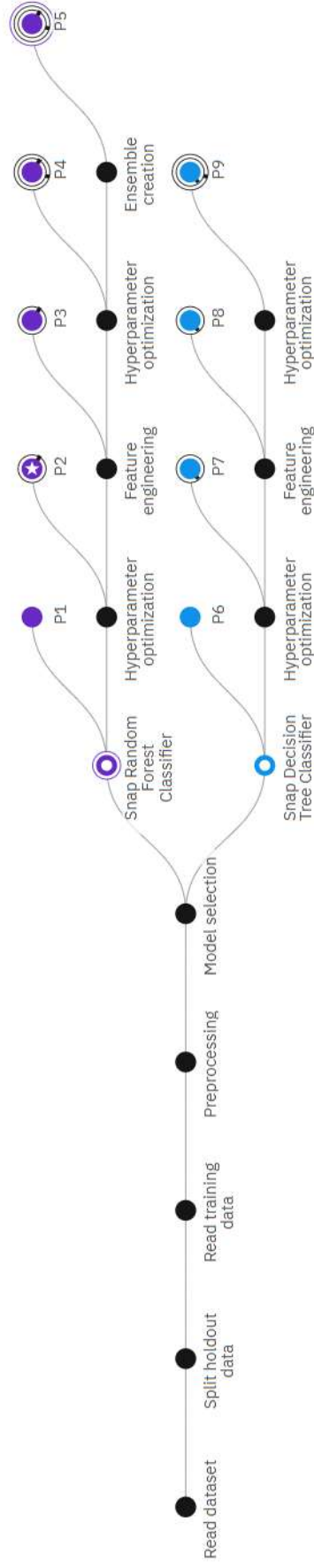
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Experiment summary

Pipeline comparison

Progress map ⁽ⁱ⁾

Prediction column: Failure Type



Pipeline leaderboard 🔍

Pipeline Comparison

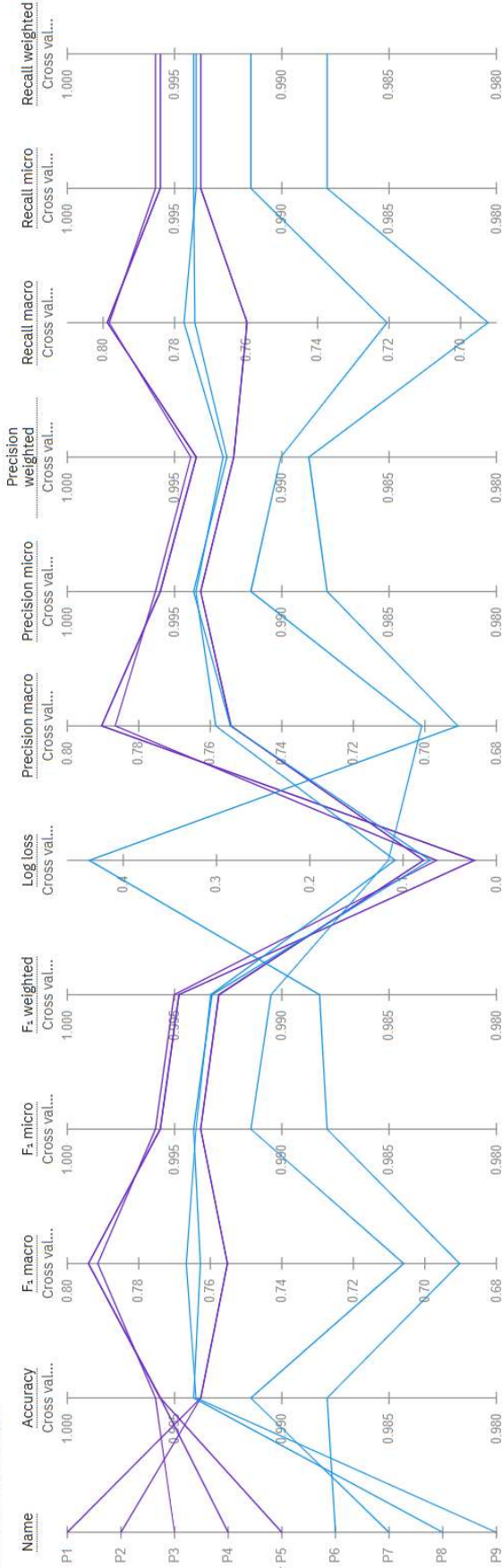
Experiment summary

Pipeline comparison

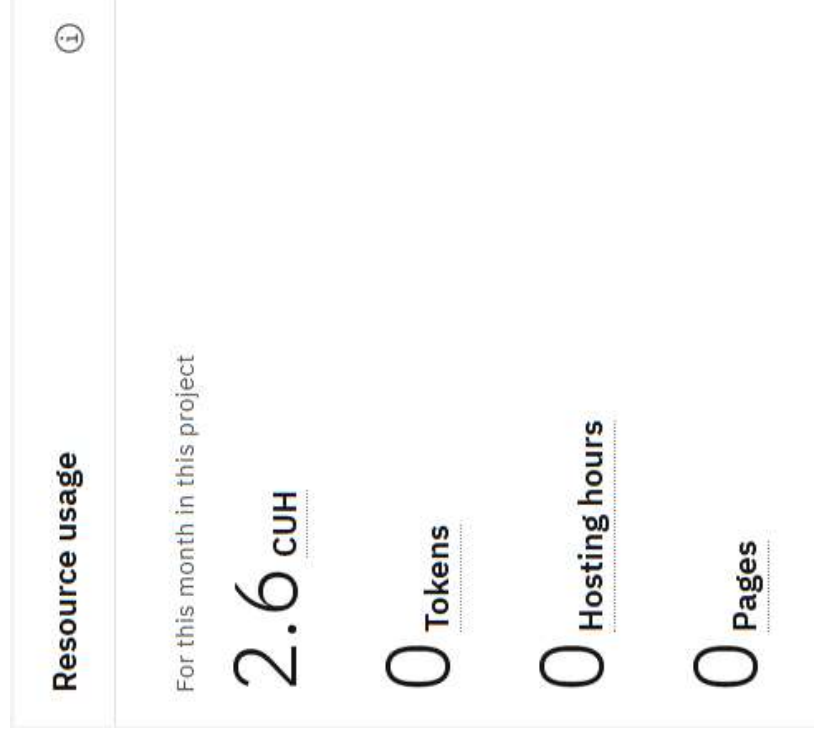
★ Rank by: F₁ macro (Optimized) | Cross validation score

Metric chart ⓘ

Prediction column: Failure Type



Token Usage



Model Training

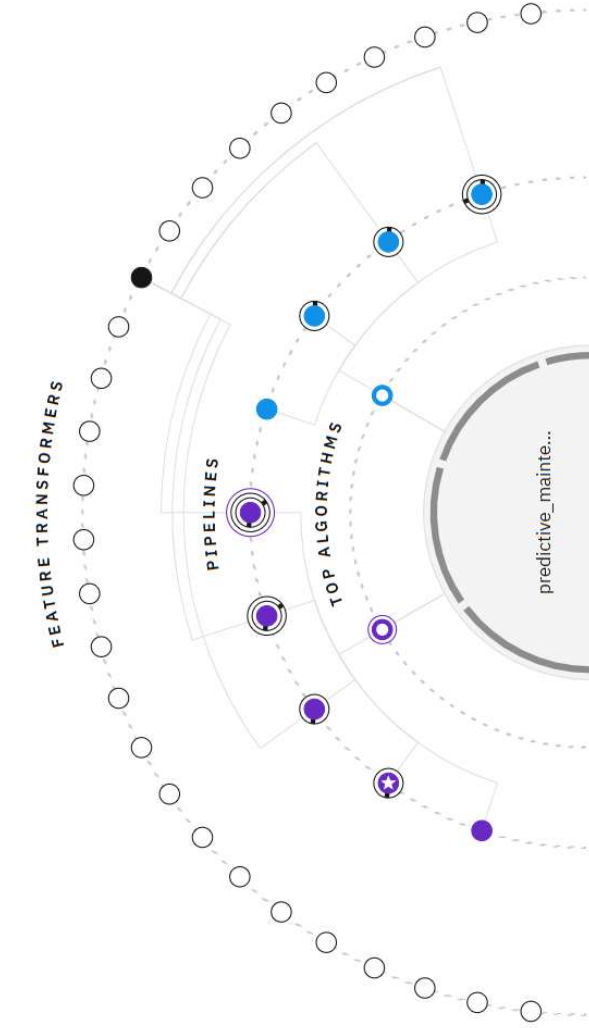
Projects / 5/6/2026-ibm-edunet / machine learning

Experiment summary

Pipeline comparison

Relationship map

Prediction column: Failure Type



Progress map

[Swap view](#)



Experiment completed

9 PIPELINES GENERATED

9 pipelines generated from algorithms. See pipeline leaderboard below for more detail.

Time elapsed: 5 minutes

[View log](#)

[Save code](#)

Pipeline leaderboard

Heat-Map

Observed	Predicted						
	Heat Dissipation Failure	No Failure	Overstrain Failure	Power Failure	Random Failures	Tool Wear Failure	Percent correct
Heat Dissipation Failure	10	0	1	0	0	0	90.9%
No Failure	0	965	0	0	0	0	100.0%
Overstrain Failure	0	0	8	0	0	0	100.0%
Power Failure	0	0	0	10	0	0	100.0%
Random Failures	0	2	0	0	0	0	0.0%
Tool Wear Failure	0	0	0	0	0	4	100.0%
Percent correct	100.0%	99.8%	88.9%	100.0%	0.0%	100.0%	99.7%

Less correct

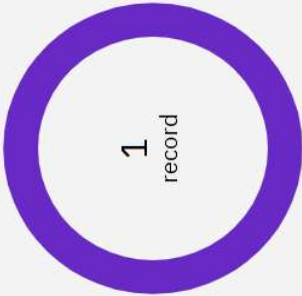
More correct

Model Testing

Prediction results

Prediction type
Multiclass classification

Prediction percentage



No Failure

Confidence level distribution

Display format for prediction results
☒ Table view ☐ JSON view

Show input data ⓘ

	Prediction	Confidence	UDI	Product ID	Type
1	No Failure	73%	1	c_1	L
2					
3					
4					
5					
6					
7					
8					
9					
10					

Download JSON file

Role of Agentic AI in the solution

Role of Machine Learning in Predictive Maintenance

Machine Learning enables the system to function as an intelligent, predictive engine rather than relying solely on manual, reactive alarms. Using IBM watsonx.ai, the model identifies complex, hidden patterns in sensor data that precede physical failures.

- **Multi-Dimensional Analysis:** The ML model (Snap Random Forest Classifier) understands how multiple variables—such as how torque interacts with rotational speed and temperature simultaneously—affect the machine's health in real-time.
- **Catching Rare Anomalies:** By optimizing for the F1 Macro score, the AI effectively handles the extreme class imbalance in industrial data (96.5% healthy vs. 3.5% failures), ensuring it catches rare events like Tool Wear or Heat Dissipation failures instead of just guessing "Healthy."
- **Context-Aware Diagnostics:** Instead of just triggering a generic "Error," the ML model categorizes the exact type of failure and outputs a confidence-scored diagnostic verdict. Overall, Machine Learning makes the system proactive and safe, acting like a smart digital diagnostic technician that helps operators maintain optimal uptime.



Technology Used

- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **IBM watsonx.ai (AutoAI)** – This is the core technology shown in your picture. It is the IBM Cloud service you used to automatically train, evaluate, and deploy your predictive maintenance model.
- **Streamlit** – The framework you used to build the interactive web dashboard for your app.
- **Python** – The programming language used to connect your Streamlit app to the IBM watsonx.ai prediction endpoint via REST API.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database (FAISS/Chroma)** – For RAG-based nutritional data retrieval

Novelty and Uniqueness

The proposed Nutrition Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware nutrition guidance.

RAG-Based Accuracy – Integrates real-time data retrieval with AI generation, reducing misinformation and ensuring trustworthy recommendations.

Hyper-Personalization – Tailors diet plans based on health conditions, cultural preferences, and fitness goals.

Real-Time Feedback Loop – Continuously adapts recommendations based on daily food logs and user behavior.

Multi-Modal Input – Supports text, voice, and image-based meal tracking for better usability.

Preventive Healthcare Focus – Goes beyond diet planning to provide disease-specific nutrition advice.
Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital nutritionist**, not just a static diet recommendation tool.

Git Hub Link

Git Hub Link : <https://github.com/Basheer78654/streamlit-app.git>

Streamlit App link : <https://app-app-86cuhmm55zzarihythmpxo.streamlit.app/>

Future Scope

- **1) Data Ingestion & Splitting:** AutoAI reads the raw dataset and automatically sets aside a "holdout" portion of the data to test the model fairly later.
- **2) Preprocessing & Model Selection:** The engine cleans the data and rapidly tests multiple classification algorithms (like Decision Trees and Random Forests) to see which handles the data best.
- **3) Optimization (FE & HPO):** AutoAI applies automated Feature Engineering (FE) and Hyperparameter Optimization (HPO) to the winning models to maximize their F1 Macro scores.
- **4) Final Pipeline Deployment: Pipeline 3 (Snap Random Forest)** emerged as the winner and was automatically deployed as a secure API endpoint for our Streamlit app to use.

Thank You!

Thank you for your time and interest.